

Biased Samples and Mismatched Denominators

Answer Key

Grade 7

Quick Answers

- | | |
|--|---|
| 1. (a) the corrected probability = $\frac{5}{12}$, — | 6. (a) the corrected probability = $\frac{9}{19}$, — |
| 2. (a) the mean of the sample = 95, — | 7. (a) the corrected probability = $\frac{2}{5}$, — |
| 3. (a) the corrected probability = $1 \cdot \frac{1}{3}$, — | 8. (a) the number the organiser calculated = |
| 4. (a) the corrected probability = $\frac{2}{3}$, — | 212.5, — |
| 5. (a) the estimate the survey gives = 15000, — | |
-

What is verified

0 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 1, 2, 3, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 7

7.SP.A – *problems 2, 5, 8*

7.SP.C.5 – *problems 1, 7*

7.SP.C.7 – *problems 3, 4, 6*

Difficulty 1–5 of 5 across 17 tagged checks.

- 1.** Bag of 5 red and 7 blue; Sam gives the probability of red as $\frac{5}{7}$.

Solution: (a) The bag holds $5 + 7 = 12$ counters and every one is equally likely, so the probability of red is $\frac{5}{12}$. $\frac{5}{12}$

(b) **Instructor-judged.** The denominator of a probability is the total number of equally likely outcomes. Sam used 7, the number of blue counters, which is not a total of anything the draw chooses from. Full credit needs the words “total number of counters”, or an equivalent statement that the bottom of the fraction must count every counter in the bag.

- 2.** Homework times of the 10 students who stayed late in the library.

Solution: (a) The ten times total 950 minutes, and $950 \div 10 = 95$ minutes. The arithmetic is correct. 95

(b) **Instructor-judged.** The sample is not random. Only students who stayed late in the library were asked, and those students are more likely than average to be doing a lot of homework, so 95 minutes is almost certainly too high for the school as a whole. Full credit needs the reason the group was unusual — not just the word “biased” — and ideally the direction of the error.

3. 4 green and 6 yellow sweets; one green is eaten; Ana keeps $\frac{4}{10}$.

Solution: (a) Eating a green sweet changes *both* counts: 3 green are left out of 9 sweets, so the probability is $\frac{3}{9} = \frac{1}{3}$. $\frac{1}{3}$

(b) **Instructor-judged.** Ana updated neither number. The green sweet that was eaten comes off the top (4 becomes 3) and off the bottom (10 becomes 9). Full credit needs both changes; noticing only that the total fell is half the answer.

4. Spinner of 12 sections (4 red, 3 blue, 5 green); Ben gives “not red” as $\frac{4}{12}$.

Solution: (a) Not red means blue or green, which is $3+5 = 8$ of the 12 sections: $\frac{8}{12} = \frac{2}{3}$. $\frac{2}{3}$

(b) **Instructor-judged.** Ben’s denominator of 12 is right and his numerator is not: he counted the red sections, which are exactly the outcomes the question excludes. Full credit needs the correct favourable count of 8, or the equivalent route $1 - \frac{4}{12} = \frac{2}{3}$.

5. 40 people surveyed leaving a cinema, 30 go weekly; town of 20000.

Solution: (a) The sample fraction is $\frac{30}{40} = 0.75$, so the survey estimates $0.75 \times 20000 = 15000$ residents. 15000

(b) **Instructor-judged.** Everybody in the sample was already at the cinema, so the sample was selected by the very behaviour being measured, and a resident who never goes had no chance at all of being asked. The estimate of 15000 is therefore far too high. Full credit needs the self-selecting nature of the group plus the direction of the error.

6. Cards 1 to 20 with the 7 removed; Priya gives odd as $\frac{10}{20}$.

Solution: (a) There were 10 odd cards among 20. The card removed, the 7, is odd, so 9 odd cards remain among 19: $\frac{9}{19}$. $\frac{9}{19}$

(b) **Instructor-judged.** Priya used the counts from before the card was removed. Because the removed card was itself one of the favourable outcomes, it comes off the top as well as the bottom. Full credit needs that point; a response that only reduces the total to 19 has half of it.

7. Class of 30; 18 play an instrument; 12 both play and sing. Leo gives $\frac{12}{18}$.

Solution: (a) The student is chosen from the whole class, so the denominator is 30:
 $\frac{12}{30} = \frac{2}{5}$. $\frac{2}{5}$

(b) **Instructor-judged.** The denominator must match the group the choice is made from. Leo divided by the 18 instrument players, which answers a different question — the chance that a *player* also sings — rather than the chance that a member of the class does both. Full credit needs that mismatch named.

8. Club of 250; 40 of them reply and 34 say yes.

Solution: (a) $\frac{34}{40} \times 250 = 0.85 \times 250 = 212.5$, so the organiser announced about 213 members. 212.5

(b) **Instructor-judged.** The 40 repliers chose themselves. Members who are keen on the trip are the ones most likely to answer an email about it, so the fraction saying yes among repliers is higher than among all 250 members, and the estimate is too high. Full credit needs the self-selection point plus the direction of the error.

(c) **Instructor-judged.** Any method earns credit that gives every one of the 250 members the same chance of being counted — for example, numbering all 250 members, drawing a sample at random, and then following those particular members up until they answer, instead of accepting whoever happens to reply. Emailing everyone a second time does not earn credit: it repeats the same self-selection.
